Carole Ibrahim

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LinkedIn: linkedin.com/in/carole-ibrahim-049883160

GitHub: github.com/carole-ibrahim

Personal website: carole-ibrahim.github.io

Technical Skills

Programming Languages: Python, Bash, Scala, Java
AI Frameworks: PyTorch, TensorFlow, Transformers, Langchain, Sentence Transformers
Agentic: RAG, GraphRAG, AI Agents, MCP Servers
MLOps: MLflow, Airflow, Docker, Kubernetes, ArgoCD, Terraform, helm, GitHub Actions, FastAPI, Streamlit
Data Engineering: SQL, NoSQL, DBT, ETL/ELT, Vector Stores
Cloud Platforms: GCP, Azure, Scaleway
AI Domains: NLP, Time-Series, Tabular data, Recommender Systems, RL

Work Experience

Data Scientist *Nov 2024 – Present*
pass Culture

- Published a research paper at **RecSys 2025** Workshop in collaboration with INRIA Scalay.
- Enhanced recommendation pipeline using Two-Tower models, multi-armed bandits, and determinantal point processes.
- Developed an AI Agent using LLMs and **LangChain** for automated content conformity checks.
- Maintained and optimized production **infrastructure** on GCP (Vertex AI, Cloud Run, Airflow, k8s clusters).

Machine Learning Engineer *Apr 2023 – Oct 2024*
Lusis

- Developed and benchmarked ML models (decision trees, transformers, neural networks) for a FOREX hedge fund.
- Benchmarked Informer, Autoformer, and Transformer architectures on financial time series using Jean-Zay supercomputer using **SLURM** parallel computing. Work published as a **CIR** publication.

Deep Learning Researcher *Feb 2023 – Apr 2023*
Conservatoire National des Arts et Métiers (CNAM)

- Trained an **OCR** model for handwritten text using **CNN** + **LSTM** architecture.

Data Scientist *Jun 2022 – Sep 2022*
Ericsson

- Built **NLP** pipelines using **LDA**, **TF-IDF**, and n-grams to classify logs technical documents.

Research Publications

- **C. Ibrahim** et al. *"Diversified Recommendations of Cultural Activities with Personalized Determinantal Point Processes."* RecSoGood Workshop at RecSys 2025, ACM, 2025. arXiv:2509.10392

Education

Institut Polytechnique de Paris *2022 – 2023*
Master's in Artificial Intelligence and Data Analysis (GPA: 4.0/4.0, Rank: 1/32)

Télécom SudParis *2020 – 2023*
Master's in Computer Science and Artificial Intelligence (GPA: 3.78/4.0)

Lycée Fénelon *2017 – 2020*
*French Preparatory Classes MPSI/MP**

Certifications

- AI Agents Fundamentals – Hugging Face.
- Azure AI Engineer Associate – Microsoft Azure.
- Machine Learning Nanodegree – Udacity.
- Advanced Data Analysis Nanodegree – Udacity.

Languages

French (C2), English (C2), Arabic (C2), Spanish (B2)

Awards & Leadership

- Awarded a 5-year scholarship from the French Government for academic excellence.
- Vice President, INTECH Robotics Club — led team participation in the French Robotics Cup (2020–2022).